Ahmad Bashir

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Professional Summary

Senior Machine Learning Engineer specialising in **generative video** — diffusion models, avatar synthesis, and lip-sync — with 3.5 years shipping production ML at Colossyan (AI video generation platform). Experience across the full model lifecycle: large-scale data curation (**10,000+ hrs** video, Spark/Argo), model training and evaluation, and **real-time inference in production**. Recent work in agentic LLM systems and RAG.

Work Experience

Colossyan Inc. Budapest, Hungary (Remote)
Senior Machine Learning Engineer *Dec 2022 – Present*

- **Avatar Pipeline Architecture & Lip-Sync**: Redesigned the core avatar generation pipeline, improving **lip-sync accuracy** and avatar fidelity by **20%** in production.
- **Large-Scale Data Processing**: Architected a distributed data pipeline (**Apache Spark**, **Argo Workflows**) that processed **10,000+ hours** of raw video through 6 frame- and video-level quality filters, producing the curated training set for production avatar models.
- **AI Avatar & Gesture Synthesis**: Built and deployed models for gesture synthesis and speech-to-video synchronisation, contributing to a **30% increase in user engagement**.
- **Generative AI R&D**: Lead diffusion-model R&D for avatar synthesis, taking techniques from paper to production deployment.

Intelligent Machines Lab, ITU Lahore, Pakistan
Research Assistant *Jun 2021 – Nov 2022*

- **3D Reconstruction & NeRF**: Developed PyTorch models for 3D reconstruction, NeRF preprocessing, and point-cloud processing, improving reconstruction accuracy by **15%** over the lab baseline.
- **Camera Pose Estimation**: Implemented camera pose estimation for 3D vision pipelines, reaching **90% accuracy** on benchmark datasets.
- **Research Publication**: Co-authored a research paper on novel 3D point cloud upsampling methods.

Selected Projects

Botvee.ai — Agentic Orchestration Platform (2025–2026): Built an **agentic AI platform** for customer support and lead generation: intent-routing engine executing multi-step workflows (CRM sync, meeting scheduling) via external APIs, and a **RAG pipeline** with confidence-based human-in-the-loop handoff — automating **85% of support volume** at **4.8/5.0 CSAT**. Tech: Python, FastAPI, OpenAI/Anthropic APIs, vector DBs, LangChain.

AI-Powered Video Interview Proctoring System (2024–2025): Built a real-time AI proctoring system integrating **deepfake detection**, gaze tracking, and facial behaviour analysis for automated interview integrity verification. Tech: Python, PyTorch, OpenCV.

Cancerous Blood Cell Detection (2024): Trained a **YOLOv11** detector for cancerous blood cells in microscopy imagery, reaching **95% detection accuracy**. Tech: Python, PyTorch, YOLO.

Real-Time Video Background Matting (2023): Optimised a real-time background-matting model, improving inference efficiency by **20%** with no visual-quality loss. Tech: Python, PyTorch, OpenCV.

Neural Radiance Fields Without COLMAP (2022): Developed a **NeRF-based 3D reconstruction** method that removes Structure-from-Motion (SfM) preprocessing entirely, enabling reconstruction of texture-poor surfaces without COLMAP. Tech: Python, PyTorch.

Technical Skills

Languages: Python, C++, SQL, TypeScript

ML & Modeling: PyTorch, diffusion models (Stable Diffusion), Transformers, NeRF / 3D reconstruction, object detection (YOLO), CLIP, Segment Anything (SAM), TensorFlow, Scikit-Learn

ML Systems: distributed data processing (Apache Spark, Argo Workflows), model deployment (ONNX, Docker, Kubernetes, AWS EC2/S3), real-time video inference (FFmpeg, OpenCV), experiment tracking (Weights & Biases, TensorBoard)

LLM Engineering: RAG pipelines, agentic workflows, LangChain, LangGraph, vector databases, FastAPI

Education

Information Technology University of the Punjab Lahore, Pakistan
BS Computer Science | Teaching Assistant, Parallel & Distributed Computing (2021–22) *Sep 2018 – Jun 2022*

Awards & Languages

- **2nd Place Nationally** – IEEE Xtreme Programming Competition (2021); **2nd Place** – STS Programming Competition, Superior University Lahore (2021); **Top 5 Finalist** – Ignite CyberSecurity Hackathon Pakistan (2022)
- **Languages**: Urdu (native) · English (C1, TOEFL iBT 103/120) · German (A2)